

Practice Questions for 10L1-CrossValidation

Generated by Scribe.AI

December 18, 2024

1 Questions

1. According to the lecture, what is the purpose of using held-out validation data?
 - A. To estimate the model parameters
 - B. To tune the model hyperparameters
 - C. To evaluate the final model performance
 - D. To train the model
 - E. To increase the size of the training data
2. In the context of regularized linear regression, what does the regularization coefficient λ control?
 - A. The residual sum of squares (RSS)
 - B. The training error
 - C. The complexity of the model
 - D. The size of the training data
 - E. The number of features in the model
3. According to the lecture, what does a learning curve with a high training accuracy and a significantly lower test accuracy indicate?
 - A. Underfitting
 - B. Overfitting
 - C. Optimal model complexity
 - D. Insufficient training data
 - E. High bias

2 Answers

1. According to the lecture, what is the purpose of using held-out validation data?
 - A. Answer A is incorrect because model parameters are estimated from the training data, not the validation data.
 - B. Answer B is correct because held-out validation data is used to tune hyperparameters, not to estimate parameters or evaluate the final model.
 - C. Answer C is incorrect because the final model performance is evaluated on the test data, not the validation data.
 - D. Answer D is incorrect because the model is trained on the training data, not the validation data.
 - E. Answer E is incorrect because held-out validation data is a separate dataset and does not increase the size of the training data.
2. In the context of regularized linear regression, what does the regularization coefficient λ control?
 - A. Answer A is incorrect because the regularization coefficient does not directly control the RSS, but rather influences it through the regularization term.
 - B. Answer B is incorrect because the regularization coefficient does not directly control the training error, but rather influences it through the regularization term.
 - C. Answer C is correct because the regularization coefficient λ controls the trade-off between training error and model complexity.
 - D. Answer D is incorrect because the regularization coefficient does not control the size of the training data.
 - E. Answer E is incorrect because the regularization coefficient does not control the number of features in the model.
3. According to the lecture, what does a learning curve with a high training accuracy and a significantly lower test accuracy indicate?
 - A. Answer A is incorrect because underfitting is characterized by low accuracy on both training and test sets.
 - B. Answer B is correct because a high training accuracy and a lower test accuracy is a classic sign of overfitting.
 - C. Answer C is incorrect because optimal model complexity would have similar performance on both training and test sets.
 - D. Answer D is incorrect because while insufficient training data can contribute to overfitting, the key indicator is the discrepancy between training and test accuracy.
 - E. Answer E is incorrect because high bias is associated with underfitting, not overfitting.